Syntax Analysis

Dynamic Programming Parsing Methods

- CYK (Cocke-Younger-Kasami, 1967) algorithm based on **bottom-up** parsing and requires first normalizing the grammar.

- Earley (1970) parser is based on **top-down** parsing, does not require normalizing the grammar, but is more complex.

CYK parsing algorithm

- First, CFG must be converted to Chomsky normal form (CNF)

  ✓ Productions must have either exactly 2 non-terminal symbols on the RHS or 1 terminal symbol.

- Parse bottom-up, storing phrases formed from all substrings in a triangular table (chart).

Conversion to Chomsky Normal Form

- Eliminate all ε rules of the form $A \rightarrow \varepsilon$ ($A \neq S$): remove rule and split rules containing A on the RHS in all versions. S is Start Symbol.

- Eliminate all unit rules $A \rightarrow B$, by adding all $B \rightarrow R_i$ to $A \rightarrow R_i$ where R_i is not a unit rule. If R_i is a unit rule add all $R_i \rightarrow K_i$ to A ($A \rightarrow K_i$) where K_i is not a unit rule.

  ✓Continue this process for all following unit rules, Then eliminate $A \rightarrow B$.

- Clean up remaining rules: For $A \rightarrow R_1, R_2, .. R_n$ ($n>2$, R_i terminals or nonterminals),

  ✓create a chain $\{A \rightarrow R_1 A_1, A_1 \rightarrow R_2 A_2 \ldots A_{n-2} \rightarrow R_{n-1} R_n\}$.

  ✓for all R_i that are terminals, create a lexicon rule and replace R_i with its LHS.

Conversion to Chomsky Normal Form

CFG

S → NP VP
S → Aux NP VP
S → VP
NP → Pronoun
NP → Proper-Noun
NP → Det Nominal
Nominal → Noun
Nominal → Nominal Noun
Nominal → Nominal PP
VP → Verb
VP → Verb NP
VP → VP PP
PP → Prep NP
Det → the | a | that | this
Noun → book | flight | meal | money
Verb → book | include | prefer
Pronoun → I | he | she | me
Proper-Noun → Houston | NWA
Aux → does
Prep → from | to | on | near | through

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CNF

S → NP VP
S → X1 VP
X1 → Aux NP
S → book | include | prefer
S → Verb NP
S → VP PP
NP → I | he | she | me
NP → Houston | NWA
NP → Det Nominal
Nominal → book | flight | meal | money
Nominal → Nominal Noun
Nominal → Nominal PP
VP → book | include | prefer
VP → Verb NP
VP → VP PP
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Det → the | a | that | this
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Quick Exercise - CFG to CNF

CFG

$$
\begin{aligned}
S &\rightarrow b\ A \mid a\ B \mid \epsilon \\
A &\rightarrow a \mid a\ S \mid b\ A\ A \\
B &\rightarrow b \mid b\ S \mid a\ B\ B
\end{aligned}
$$

Solution - CFG to CNF

CFG

$$
\begin{aligned}
S &\rightarrow b\ A \mid a\ B \mid \epsilon \\
A &\rightarrow a \mid a\ S \mid b\ A\ A \\
B &\rightarrow b \mid b\ S \mid a\ B\ B
\end{aligned}
$$

CNF

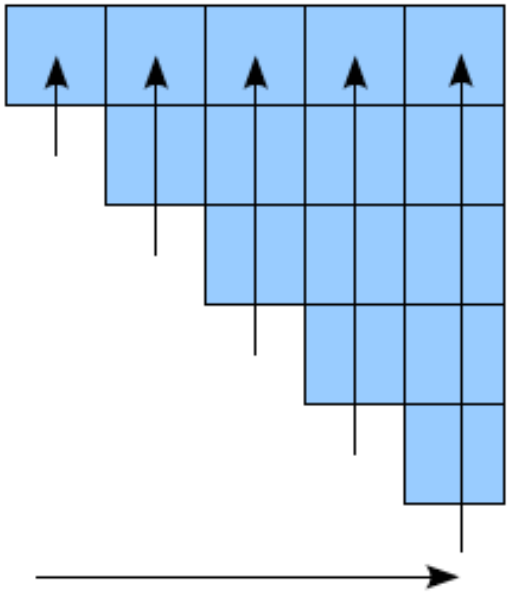
$$
\begin{aligned}
S &\rightarrow T_b\ A \mid T_a\ B \mid \epsilon \\
A &\rightarrow a \mid T_a\ S \mid T_b\ A_2 \\
A_2 &\rightarrow A\ A \\
B &\rightarrow b \mid T_b\ S \mid T_a\ B_2 \\
B_2 &\rightarrow B\ B \\
T_a &\rightarrow a \\
T_b &\rightarrow b
\end{aligned}
$$

Parse Table

Book the flight through Houston

0 Book 1 the 2 flight 3 through 4 Houston 5

Book	the	flight	through	Houston
S,VP,Verb Nominal, Noun [0,1]	[0,2]	S,VP,X2 [0,3]	[0,4]	S, VP [0,5]
	Det [1,2]	NP [1,3]	[1,4]	NP [1,5]
		Nominal, Noun [2,3]	[2,4]	Nominal [2,5]
			Prep [3,4]	PP [3,5]
				NP, Proper-Noun [4,5]



Each cell [*i, j*] in this matrix contains a set of non-terminals that represent all the constituents that span positions *i* through *j* of the input